
Solving the Lorenz-1960 ODE System Using Optimal ANN Architectures

By

Ikramul Hasan Moral (0112230489)

Md. Abu Bakar (0112230200)

Samiur Rahman Omlan (0112230195)

Fariha Islam (0112230431)

Md. Touhidul Islam (0112230435)

Submitted in partial fulfilment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

April 3, 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNITED INTERNATIONAL UNIVERSITY

Abstract

Nonlinear coupled ordinary differential equations (ODEs) occur often in physical modeling, and in most cases, their analytical solutions cannot be obtained. The orthodox method of solving such problems is numerical methods like the Runge-Kutta family and these are not free in computation and mesh dependency. That system is the Lorenz-1960 ODE system, a simplified meteorological model, which serves as a test case in this project, and poses a straightforward question: is it possible to use a plain artificial neural network (ANN), with neither physics informed loss nor operator learning configuration, to learn to approximate the solution to this coupled system? In response to this, an accurate reference solution is initially calculated using the Runge-Kutta based solver in the SciPy. There are several feedforward ANN structures that are then trained on this reference data, and the different variations are the number of hidden layers, the number of neurons within any given layer and the activation function. The models are tested using approximation error, training convergence, as well as time of computation. The purpose of this comparison is to produce the most appropriate architectural configuration of this particular ODE system and to contextualize the results with regard to the more recent literature on neural network based solvers of differential equations. The final product of this study is the possession of an empirically-based opinion regarding what plain ANNs are and are not capable of as ODE solvers.

Acknowledgements

This work would have not been possible without the input and support of many people over the last two trimesters. We would like to express my gratitude to everyone who contributed to it in some way or other.

First, we would like to thank my academic advisors, ...

Our sincere gratitude goes to ...

We are also thankful to ...

Last but not the least, We owe to our family including our parents for their unconditional love and immense emotional support.

Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

- 1.

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

Table of Contents

Table of Contents	v
List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Project Overview	1
1.2 Motivation	1
1.3 Objectives	2
1.4 Methodology	2
1.5 Project Outcome	3
1.6 Organization of the Report	4
2 Background	5
2.1 Preliminaries	5
2.2 Literature Review	5
2.2.1 Similar Applications	5
2.2.2 Related Research	5
2.3 Gap Analysis	5
2.4 Summary	5
3 Project Design	6
3.1 Requirement Analysis	6
3.1.1 Functional and Nonfunctional Requirements	6
3.1.2 Context Diagram	6
3.1.3 Data Flow Diagram Level 1	6
3.1.4 UI Design	6
3.2 Detailed Methodology and Design	6
3.3 Project Plan	6
3.4 Task Allocation	6
3.5 Summary	6

4	Implementation and Results	7
4.1	Environment Setup	7
4.2	Testing and Evaluation	7
4.3	Results and Discussion	7
4.4	Summary	7
5	Standards and Design Constraints	8
5.1	Compliance with the Standards	8
5.1.1	Software Standards	8
5.1.2	Hardware Standards	8
5.1.3	Communication Standards	8
5.2	Design Constraints	8
5.2.1	Economic Constraint	9
5.2.2	Environmental Constraint	9
5.2.3	Ethical Constraint	9
5.2.4	Health and Safety Constraint	9
5.2.5	Social Constraint	9
5.2.6	Political Constraint	9
5.2.7	Sustainability	9
5.3	Cost Analysis	9
5.4	Complex Engineering Problem	9
5.4.1	Complex Problem Solving	9
5.4.2	Engineering Activities	9
5.5	Summary	9
6	Conclusion	11
6.1	Summary	11
6.2	Limitation	11
6.3	Future Work	11
	References	12

List of Figures

1.1	Methodology Diagram	3
-----	-------------------------------	---

List of Tables

5.1	Mapping with complex problem solving.	9
5.2	Mapping with complex engineering activities.	10

Chapter 1

Introduction

This chapter introduces the research topic, provides an overview and motivation for the study, and outlines the main objectives and expected outcomes. It also outlines the methodology and briefly describes how the rest of the report is organized.

1.1 Project Overview

Ordinary Differential Equations (ODEs) describe the dynamic changes of quantities over time and are widely used in fields such as physics, engineering, and mathematics. In many real-world problems, particularly those involving complex or nonlinear equations, it is often not possible to obtain exact solutions. That is why methods such as Euler's method and Runge-Kutta are used to obtain approximate solutions. In this project, we examine a particular system consisting of three nonlinear equations known as the Lorenz-1960 system (i.e., one of the first meteorological models). The system is small but complex enough to make a good testing ground for various solution methods. The Lorenz-1960 system is a great candidate for evaluating how well ANNs learn and predict behavior. Classical numerical methods address these problems step by step and are typically accurate, but they tend to be slow and require substantial computing power. Generated data from numerical methods, such as Runge-Kutta, can be used to train ANNs to model the relationship between time and the variables. During training, the ANN learns to map its predictions to those of the numerical solver. Once the ANN has been trained, it can provide rapid predictions of outcomes for any time point without needing to solve the equations again. Therefore, the primary objective of this project is to solve the Lorenz-1960 system and to see how well a simple ANN can learn its behavior and which ANN setup yields optimal results.

1.2 Motivation

This project is motivated by both computational and research perspectives. Numerical methods are solid and accurate. While these numerical methods are well-suited for direct

simulation, they become less efficient when repeated evaluations are required. The subsequent solution of the equation can then use the ANN as a direct approximator. After training, the model outputs predictions and acts as a surrogate to the system. Thus, approximating the solution using an ANN is another approach. There is a growing body of literature applying neural approaches to differential equations (DE), including physics-informed neural networks (PINNs), operator-learning, and ANN-based approximators, among others. This discrepancy renders this issue both timely and appropriate for exploration. Another motivation is the educational and practical benefits of this project. This project provides foundations for fostered expertise in ODE solving, neural network modeling, comparative experimentation, and scientific analytical processes. This makes any such comparative evaluation valuable in itself, even if the results would indicate that simple ANNs are outperformed by more advanced methods - such findings will help clarify the limitations of plain ANN approaches.

1.3 Objectives

Our research paper primarily aims to look into whether a plain ANN learns the chaotic Lorenz 1960 ODE system. Some specific objectives are given below:

- As the Lorenz system is a three-coupled nonlinear, deterministic, and chaotic system, to solve this, we combine different areas such as ODE, numerical method (Runge Kutta & SciPy solver), nonlinear, and ANN model.
- To generate a high-accuracy numerical solution and validate the ANN's performance, compare ANN results with different numerical methods, a benchmark formulation was created using the traditional Runge-Kutta method.
- To evaluate several Feedforward ANN architectures, we must experiment with depth(number of hidden layers) and width(number of neurons per layer), and activation function to get the most efficient structure. This objective determines if the ANN structure can achieve the high accuracy provided by the Runge-Kutta method.
- The performance of the model will be analyzed by error metrics(mean squared error) to evaluate if a plain ANN can learn the Lorenz system.
- Compare results with the PINN-based result from [1] (PinnDE) and the survey-based results from [2].

1.4 Methodology

Our research will follow some stages, and in the beginning, we learn and understand Lorenz -1960 three equations. After that we are identifying which ANN architecture will approximate its solution and give the best outcome. In this methodology, we only cover

our FYDP 1 approach to solve the Lorez-1960 ODE system using ANN and identify the best optimal architecture for our research. Our research follow some stages:

To begin with, as the Lorez system is “chaotic”, meaning a small error can grow very fast and impact the whole system so it can be defined as an initial value problem.

And now this system will be solved using two methods: first, a validated numerical baseline using a high-accuracy Runge-Kutta solver in Python and second, a high-accuracy SciPy solver for reference comparison.

Next, to simplify supervised learning, first generate aligned reference dataset and then problem framing for learning. This resulting dataset ensures optimization against numerical baseline.

Then comes the ANN architecture design based on literature-guided parameters to ensure optimal performance. Here a family of feedforward ANN mode will be constructed based on depth, width, activation function, and optimization strategy.

After that to maintain computational efficiency on standard hardware while preserving the system’s chaotic dynamics, the data is densely sampled and partitioned into training, validation, and testing subsets. This follows the standard data-driven modeling approach explored in recent ANN surveys for differential equations [2].

And now compare against the numerical baseline. It means taking your trained ANN models and measuring how well they perform against two distinct reference points.

Lastly, comparing all ANN models against the numerical and literature based benchmarks considering factors such as structural simplicity, training stability, and reproducibility. We analyze the trade-offs among them. Which architecture will satisfy all this then it will be a recommended solution for the Lorez-1960 benchmark.

For better understanding diagram of our methodology part is given below:-

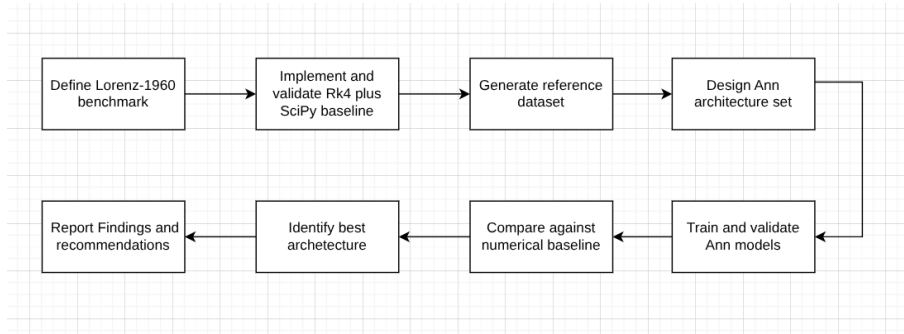


Figure 1.1: Methodology Diagram

1.5 Project Outcome

The expected outcome of the project will be the clear and the justified determination of the most appropriate architecture for the ANN in estimating the solution to the Lorez-1960 system of Ordinary Differential Equations (ODEs) within the scope of the experiment.

This project must also include a numerical reference solution to the ODEs, the establishment of ANN baselines, and a comparative analysis of the results.

The project will also contribute to the current understanding of the aptitude and limitations of ANN models in the approximation and solution of differential equations. If the ANN models are able to get a satisfactory level of accuracy in approximating the solution to the ODEs, the project will have demonstrated the capability of a simple ANN architecture to effectively act as an alternative numerical solver. This would suggest that a plain ANN, without the need for physics-informed loss terms or traditional mesh-dependent numerical methods, can on its own approximate ODE solutions with sufficient accuracy while being computationally less expensive in terms of inference time and the avoidance of iterative re-solving for each query. But if the models cannot achieve satisfactory accuracy in approximating the solution to the ODEs then the study will clearly outline the constraints of simple ANN architectures and underscore the need for advanced methods like PINNs.

The expected outcome of the project will therefore be more than just the coding and experiment itself but will also involve a research based understanding of how ANN architectural choices like depth, width, and activation function can impact solution accuracy for coupled nonlinear ODE systems.

1.6 Organization of the Report

The rest of this report is outlined as follows. Chapter 2 outlines the theoretical background and literature review on Ordinary Differential Equations (ODE) solving, artificial neural networks, and existing neural methods for differential equations. Chapter 3 discusses about the project design, requirements, methodological plan, and task assignment. Chapter 4 outlines the implementation & experiment design used in this research. Chapter 5 outlines the results and analysis on the tested ANN structures. Finally, Chapter 6 concludes this report by focusing the main results, limitations, and potential directions for the future research.

Chapter 2

Background

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

2.1 Preliminaries

In this section, you have to provide the necessary background knowledge to understand the rest of the report [?].

2.2 Literature Review

This section will contain you literature review.

2.2.1 Similar Applications

Put a summary of similar web applications, mobile apps similar to your work.

2.2.2 Related Research

Here report the summary of the investigation of the research literature.

2.3 Gap Analysis

Here summarise the gap where you intend to work.

2.4 Summary

Chapter 3

Project Design

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

3.1.2 Context Diagram

3.1.3 Data Flow Diagram Level 1

3.1.4 UI Design

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint**5.2.2 Environmental Constraint****5.2.3 Ethical Constraint****5.2.4 Health and Safety Constraint****5.2.5 Social Constraint****5.2.6 Political Constraint****5.2.7 Sustainability****5.3 Cost Analysis**

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem**5.4.1 Complex Problem Solving**

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowl- edge	P2 Range of Con- flicting Require- ments	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stake- holder Involve- ment	P7 Inter- dependence
√	√					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

5.5 Summary

Table 5.2: Mapping with complex engineering activities.

A1 Range of re- sources	A2 Level of Interac- tion	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓	✓			

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

References

- [1] Jason Matthews and Alexander Bihlo. Pinnde: Physics-informed neural networks for solving differential equations, 08 2024.
- [2] Danang Pratama, Maharani Abu Bakar, Mustafa Man, and M. Mashuri. Anns-based method for solving partial differential equations : A survey, 02 2021.